

BIOGRAPHICAL SKETCH

NAME: Michael Sören Balzer, MD, FASN

AFFILIATION: Kiel University and University Hospital Schleswig-Holstein (UKSH)

POSITION TITLE: Associate Professor, Group Leader, Attending Physician

EDUCATION/TRAINING/ACADEMIC APPOINTMENTS

INSTITUTION AND LOCATION	DEGREE (if applicable)	Completion Date	FIELD OF STUDY
University of Music Karlsruhe, Karlsruhe (DE)	Junior undergrad	03/2001	Violoncello & Piano
University of Music Würzburg, Würzburg (DE)	Junior undergrad	03/2005	Violoncello & Piano
University of British Columbia, Vancouver (CA)		12/2010	Med School (Yr6), Nephrology
University College London, London (GB)		01/2011	Med School (Yr6), Pediatric Nephrology
University of Oxford, Oxford (GB)		03/2011	Med School (Yr6), Pediatrics
University of Berne, Berne (CH)		07/2011	Med School (Yr6), Surgery
Friedrich-Alexander University Erlangen-Nuremberg, Erlangen (DE)	MD	06/2012	Med School
Hannover Medical School, Hannover (DE)	Postdoctoral	10/2019	Residency, Internal Medicine; Fellowship, Nephrology
University of Pennsylvania, Philadelphia (US)	Postdoctoral	10/2022	Experimental Nephrology
Charité – Universitätsmedizin Berlin, Berlin (DE)	Habilitation	2025	Internal Medicine and Nephrology
Kiel University, Kiel (DE)	Associate Professor		Translational Nephrology
Friedrich-Alexander University Erlangen-Nuremberg, Nuremberg (DE)	MHBA (ongoing)	2027 (expected)	Health Business Administration

A. Personal Statement

I am Associate Professor (W2, tenure track) of Translational Molecular Nephrology and Single-Cell Phenotyping at Kiel University, a research group leader, and an attending physician at University Hospital Schleswig-Holstein (UKSH), with an academic affiliation at Charité. My research asks why kidney injury resolves in some patients but progresses to chronic inflammation and fibrosis in others. I combine single-cell and spatial multi-omics with human clinical phenotyping and perturbation studies to investigate adaptive and maladaptive repair and identify opportunities to prevent progression from acute kidney injury (AKI) to chronic kidney disease (CKD).

Following training at Hannover Medical School and a three-year postdoctoral fellowship in Katalin Susztak's laboratory at the University of Pennsylvania, I established my independent research group at Charité in 2022 and moved to Kiel in May 2026. I have secured approximately €2.2 million in competitive research grants and fellowships, including the €1.5 million ERC Starting Grant SINGuLAR, DFG funding, and an Else Kröner Memorial Fellowship.

Board-certified in Internal Medicine and Nephrology since 2024, my clinical interests include AKI, the AKI-to-CKD transition, and peritoneal dialysis. In Kiel, I founded and lead a post-AKI outpatient clinic, using hospital information system screening and automated alerts to identify patients at risk of being lost to follow-up. I lead an interdisciplinary research team and supervise MD, PhD, and master's projects. My ongoing MHBA studies at Friedrich-Alexander University Erlangen-Nuremberg complement this work with training in health-care management.

Selected awarded research funding

1. European Research Council (ERC) Starting Grant (SINGuLAR)
Balzer (PI), 2026–2031, €1,500,000
Mechanisms of kidney regeneration and adaptive repair
2. Clinician Scientist Program Fellowship, Berlin Institute of Health (BIH)
Balzer (PI), 2023–2025, €59,283
Cellular trajectories of repair and fibrosis following AKI
3. BIH Single Cell Approaches for Personalized Medicine Grant
Balzer (Co-PI), Halbritter (Co-PI), Kaminski (Co-PI), 2025, €30,000
Single-cell personalized medicine for ADPKD leveraging a founder mutation
4. Dr. Werner Jackstädt Research Grant
Balzer, 2024–2026, €79,661
5. Else Kröner Memorial Fellowship
Balzer (PI), 2022–2024, €230,000
Endophenotypes distinguishing adaptive and maladaptive regeneration in AKI
6. German Research Foundation (Deutsche Forschungsgemeinschaft, DFG) Grant Ba 6205/2-1
Balzer (PI), 2019–2022, €172,199
Single-cell mechanisms of renal regeneration and fibrosis in CKD

Selected Publications:

1. Miao Z.*, **Balzer, M.S.*** et al. (2021). Single cell regulatory landscape of the mouse kidney highlights cellular differentiation programs and disease targets. *Nat Commun.* doi:[10.1038/s41467-021-22266-1](https://doi.org/10.1038/s41467-021-22266-1)
2. **Balzer, M.S.** et al. (2022). Single-cell analysis highlights differences in druggable pathways underlying adaptive or fibrotic kidney regeneration. *Nat Commun.* doi:[10.1038/s41467-022-31772-9](https://doi.org/10.1038/s41467-022-31772-9)
3. **Balzer, M.S.**, Rohacs, T., Susztak, K. (2022). How many cell types are in the kidney and what do they do? *Annual Review of Physiology.* doi:[10.1146/annurev-physiol-052521-121841](https://doi.org/10.1146/annurev-physiol-052521-121841)
4. **Balzer, M.S.** et al. (2023). Treatment effects of soluble guanylate cyclase modulation on diabetic kidney disease at single-cell resolution. *Cell Rep Med.* doi:[10.1016/j.xcrm.2023.100992](https://doi.org/10.1016/j.xcrm.2023.100992)
5. Abdank, K., et al., **Balzer, M.S.** (2024). A comparative scRNAseq data analysis to match mouse models with human kidney disease at the molecular level. *Nephrol Dial Transplant.* doi:[10.1093/ndt/gfae030](https://doi.org/10.1093/ndt/gfae030)
6. **Balzer, M. S.** et al. (2026). Single-cell resolution drug effects of renin-angiotensin-aldosterone blockade in ZSF1 rat diabetic kidney disease. *J Am Soc Nephrol*, doi:[10.1681/ASN.0000000995](https://doi.org/10.1681/ASN.0000000995)

B. Positions, Scientific Appointments, and Honors

Positions and Scientific Appointments

2026–Present	Associate professor (W2, tenure track) and group leader, Translational Molecular Nephrology and Single-Cell Phenotyping, Kiel University
2026–Present	Attending physician, Department of Nephrology and Hypertension, UKSH; Founder and Lead, Post-AKI outpatient clinic
2026–Present	Associated Researcher, Department of Nephrology and Medical Intensive Care, Charité
2026–Present	Editorial Board, <i>Journal of the American Society of Nephrology (JASN)</i>
2025	Habilitation and venia legendi (Privatdozent, Priv.-Doz.) for Internal Medicine and Nephrology, “Decoding phenotypes of kidney disease at the single-cell level”, Charité
2025–2026	Attending physician, Charité, Campus Virchow-Klinikum; inpatient ward leadership and supervision of junior physicians and fellows
2024–Present	Fellow of the American Society of Nephrology (FASN)
2024	Board certification in Internal Medicine and Nephrology

2023–2026	Fellow, Berlin Institute of Health (BIH), Charité Clinician Scientist Program
2023–Present	Board member, Diabetes and Metabolism commission (DGfN)
2022–2026	Research group leader, Charité
2022–2025	Editorial fellow, <i>JASN</i>
2021–Present	Member, The New York Academy of Sciences (NYAS)
2019–Present	Member, German Academic International Network (GAIN)
2019–2022	Member, Penn Program in Single Cell Biology (PPSCB)
2017–2019	Mentee, Nephrofuture mentoring program, class of 2017, DGfN
2015–Present	Member, American Society of Nephrology (ASN)
2015–2025	Member, Junge Niere (Young Kidney), DGfN
2015–Present	Member, European Renal Association (ERA)
2015–2025	Member, ERA Young Nephrologists' Platform
2014–Present	Member, DGfN
2014–Present	Member, International Society of Nephrology (ISN)
2014–Present	Member, International Society for Peritoneal Dialysis (ISPD)
2009–Present	Honorary member, Green Templeton College, Oxford

Honors

2026	ERA Stanley Shaldon Award for Young Investigators in Translational Science
2023	DGfN Dr. Werner Jackstädt Research Prize for scientific excellence in AKI research
2015–2026	13 presentation and abstract awards (DGfN, ERA, Penn Genetics, Penn IDOM)
2010–2021	11 travel grants and congress scholarships (Keystone, DGfN, ERA, German National Academic Foundation)

C. Contributions to Science

1. Kidney cell states and treatment responses in injury and repair

My research examines why kidney injury is followed by recovery in some settings and persistent inflammation and fibrosis in others. I led single-cell studies of adaptive and fibrotic regeneration after AKI and of cellular responses to soluble guanylate cyclase modulation and renin-angiotensin-aldosterone system blockade in experimental diabetic kidney disease. These studies characterize injury-associated cell states and treatment-responsive pathways, including candidate mechanisms of maladaptive repair. The findings inform my work on the AKI-to-CKD transition and the identification of therapeutic targets for preventing progression.

- Balzer, M. S. et al. (2026). Single-cell resolution drug effects of renin-angiotensin-aldosterone blockade in ZSF1 rat diabetic kidney disease. *J Am Soc Nephrol*, doi:[10.1681/ASN.0000000995](https://doi.org/10.1681/ASN.0000000995)
- Balzer, M. S. et al. (2023). Treatment effects of soluble guanylate cyclase modulation on diabetic kidney disease at single-cell resolution. *Cell Rep Med*, doi:[10.1016/j.xcrm.2023.100992](https://doi.org/10.1016/j.xcrm.2023.100992)
- Doke, T. et al. (2022). Single-cell analysis identifies the interaction of altered renal tubules with basophils orchestrating kidney fibrosis. *Nat Immunol*, doi:[10.1038/s41590-022-01200-7](https://doi.org/10.1038/s41590-022-01200-7)
- Balzer, M. S., Rohacs, T. & Susztak, K. (2022). How Many Cell Types Are in the Kidney and What Do They Do? *Annu Rev Physiol*, doi:[10.1146/annurev-physiol-052521-121841](https://doi.org/10.1146/annurev-physiol-052521-121841)
- Balzer, M. S. et al. (2022). Single-cell analysis highlights differences in druggable pathways underlying adaptive or fibrotic kidney regeneration. *Nat Commun*, doi:[10.1038/s41467-022-31772-9](https://doi.org/10.1038/s41467-022-31772-9)
- Miao, Z.* Balzer, M. S.* et al. (2021). Single cell regulatory landscape of the mouse kidney highlights cellular differentiation programs and disease targets. *Nat Commun*, doi:[10.1038/s41467-021-22266-1](https://doi.org/10.1038/s41467-021-22266-1)
- Dhillon, P. et al. (2021). The Nuclear Receptor ESRRA Protects from Kidney Disease by Coupling Metabolism and Differentiation. *Cell Metab*, doi:[10.1016/j.cmet.2020.11.011](https://doi.org/10.1016/j.cmet.2020.11.011)
- Balzer, M. S., Ma, Z., Zhou, J., Abedini, A. & Susztak, K. (2021). How to Get Started with Single Cell RNA Sequencing Data Analysis. *J Am Soc Nephrol*, doi:[10.1681/ASN.2020121742](https://doi.org/10.1681/ASN.2020121742)
- Abedini, A. et al. (2021). Urinary Single-Cell Profiling Captures the Cellular Diversity of the Kidney. *J Am Soc Nephrol*, doi:[10.1681/ASN.2020050757](https://doi.org/10.1681/ASN.2020050757)

2. Open single-cell atlases and kidney disease research resources

I have contributed to interactive single-cell atlases spanning mouse, rat, and human kidney disease. These publicly accessible resources allow researchers to explore gene expression across cell types, disease states, and treatment conditions. Cross-species comparisons support the selection of experimental models that reflect relevant molecular features of human disease and help prioritize candidate pathways for further study. Making these datasets and analysis resources available supports reproducibility and collaborative research.

- a. Developing mouse kidney scRNA-seq & snATAC-seq atlas: [Explore atlas](#)
- b. Mouse kidney ischemia-reperfusion injury scRNA-seq atlas: [Explore atlas](#)
- c. Mouse kidney fibrosis (UUO) scRNA-seq atlas: [Explore atlas](#)
- d. Rat diabetic kidney disease snRNA-seq atlas: [Explore atlas](#)
- e. Human kidney spatial atlas: [Explore atlas](#)
- f. Mouse cross-model kidney disease single-cell atlas: [Explore atlas](#)
- g. Multispecies-integrated single-cell kidney atlas: [Explore atlas](#)

3. Multi-omic and spatial analysis of kidney disease

As co-first author, I contributed to mapping gene-expression and chromatin-accessibility patterns in the developing mouse kidney. In collaborative studies, I contributed to single-cell, multi-omic, and spatial analyses of human kidney disease and to research on urinary kidney-cell profiling. These studies relate cell states to their regulatory programs and tissue context. My last-author work comparing mouse models with human kidney disease supports the selection of experimental systems with relevant molecular features. Urinary cell profiling offers a way to study kidney-cell states without tissue biopsy; its clinical utility requires prospective validation.

- a. Klötzer, K. A. et al. (2025). Analysis of individual patient pathway coordination in a cross-species single-cell kidney atlas. *Nat Genet*, doi:[10.1038/s41588-025-02285-0](#)
- b. Abedini, A. et al. (2024). Single-cell transcriptomics and chromatin accessibility profiling elucidate the kidney-protective mechanism of mineralocorticoid receptor antagonists. *J Clin Invest*, doi:[10.1172/JCI157165](#)
- c. Abedini, A. et al. (2024). Single-cell multi-omic and spatial profiling of human kidneys implicates the fibrotic microenvironment in kidney disease progression. *Nat Genet*, doi:[10.1038/s41588-024-01802-x](#)
- d. Abdank, K. et al. (2024). A comparative scRNAseq data analysis to match mouse models with human kidney disease at the molecular level. *Nephrol Dial Transplant*, doi:[10.1093/ndt/gfae030](#)
- e. Zhou, J. et al. (2023). Unified Mouse and Human Kidney Single-Cell Expression Atlas Reveal Commonalities and Differences in Disease States. *J Am Soc Nephrol*, doi:[10.1681/ASN.0000000000000217](#)

Complete List of Published Work

<https://pubmed.ncbi.nlm.nih.gov/?term=%28Balzer+MS%29OR%28S+Balzer+M%29&sort=date>